

Problem 11.17

The annual effective interest rate for year t is

$$i_t = \frac{1}{10 + t}.$$

Calculate the current value at the end of year 2 of a payment of 6,000 made at the end of year 7.

Solution

To move the payment from the end of year 7 to the end of year 2, we must discount it using the interest rates for years 3, 4, 5, 6, and 7. The discount factor for year t is

$$v_t = \frac{1}{1 + i_t} = \frac{1}{1 + \frac{1}{10+t}} = \frac{10 + t}{11 + t}.$$

Therefore, the required current value is

$$\begin{aligned} V_2 &= 6,000 \prod_{t=3}^7 v_t \\ &= 6,000 \prod_{t=3}^7 \frac{10 + t}{11 + t} \\ &= 6,000 \left(\frac{13}{14} \right) \left(\frac{14}{15} \right) \left(\frac{15}{16} \right) \left(\frac{16}{17} \right) \left(\frac{17}{18} \right) \\ &= 6,000 \left(\frac{13}{18} \right) \\ &= 4,333.33. \end{aligned}$$

Thus, the current value of the payment at the end of year 2 is

$$\boxed{V_2 = 4,333.33}.$$